



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING

MODULE - 1

SESSION / ASSIGNMENT – 3

TASKS :

1. Write the full form and one-line use of : HTTP, HTTPS, FTP, SMTP.

➔ **HTTP – HyperText Transfer Protocol**

Used for transferring web pages and data between a web browser and a web server.

HTTPS – HyperText Transfer Protocol Secure

Used for secure communication between a web browser and a web server using encryption.

FTP – File Transfer Protocol

Used for transferring files between computers over a network.

SMTP – Simple Mail Transfer Protocol

Used for sending emails from one mail server to another.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

➔ **IPv4** uses 32-bit addresses and supports approximately 4.3 billion unique IP addresses.

IPv6 uses 128-bit addresses and supports a vastly larger number of unique IP addresses.

3. Run nslookup google.com on your system and write down the IP address you get.

➔ Name: google.com

Addresses: 2404:6800:4000:1017::8a

2404:6800:4000:1017::65

2404:6800:4000:1017::71

2404:6800:4000:1017::66

192.178.193.102

192.178.193.139

192.178.193.100

192.178.193.138

192.178.193.113

192.178.193.101

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

➔ **200** – OK

The Request was Successful

301 – Moved Permanently

The resource has been permanently moved to a new URL

404 – Not Found

The requested resource could not be found

500 – Internal Server Error

The server encountered an unexpected error

403 – Forbidden

Access to the requested resource is denied

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

➔ **DNS (Domain Name System)** is a system that translates human-readable domain names (such as google.com) into IP addresses that computers use to identify and communicate with each other over the internet.

It is also a system used to organize and locate websites on the internet.

It consists of **four main levels** (ranked based on DNS hierarchy high level to low level):

1. **Root (.)** – The highest level in the DNS hierarchy, represented by a dot (.), which is usually hidden in web addresses.
2. **TLD (Top Level Domain)** – Indicates the domain category and extension of a domain name.
The most common TLD is **com**
3. **Domain** – The main name registered by an organization.
4. **Subdomain** – A subdivision of a domain used to organize services.

Example Using www.google.com :

1. Root Domain (usually hidden) = .
2. Top-Level Domain (TLD) = **com**
3. Domain = **google**
4. Subdomain = **www**